A Survey of Commonsense Reasoning in LLMs

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Commonsense reasoning plays a crucial role in natural language processing by enabling systems to determine causation and assess the plausibility of various scenarios. Despite significant research focused on formalizing commonsense knowledge and integrating it with mathematical logic, replicating such reasoning in artificial intelligence systems—such as large language models (LLMs)—remains a formidable challenge. Despite these challenges, LLMs demonstrate significant promise and exhibit ongoing progress in reasoning capabilities. This paper surveys the current state of commonsense reasoning in LLMs, including datasets, models, benchmarks, enhancements, opportunities, and challenges—highlighting both advancements and limitations. We discuss the role of commonsense reasoning in LLMs, how they strive to emulate human-like reasoning, and why they often struggle to capture the nuanced inferences associated with commonsense understanding of the physical and social world. This survey emphasizes the need for continued investigation and optimization, including the integration of mechanisms to enhance the reliability of these models. Overall, while LLMs are making strides in commonsense reasoning, overcoming their limitations requires further research and development to fully harness their potential.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → *Artificial intelligence*; Philosophical/theoretical foundations of artificial intelligence; • **Theory of computation** → Logic.

Additional Key Words and Phrases: llms, commonsense reasoning

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1 Introduction

In philosophy, commonsense reasoning refers to drawing conclusions about the world using everyday knowledge and experience. It involves applying shared assumptions and everyday knowledge that people accept as true. This kind of reasoning is frequently implicit and is based on cultural norms and experiences that are accepted as normal in day-to-day living. There are many different domains in which commonsense is applicable, such as social commonsense, physical commonsense, biological commonsense, and many more. Since commonsense reasoning is commonly acknowledged to be an essential component of everyday decision-making and real-world situations, it is a necessary precondition for language models.

Reasoning involves deriving new conclusions from existing information and contextual cues [70, 204]. It is a fundamental component of human intelligence that supports rational thought, problem

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solving, and decision making [68, 105]. In artificial intelligence, improving reasoning has long been a central objective [142]. From a logical perspective, reasoning can be described in terms of premises, inferences, and conclusions [27]. Premises and conclusions are statements that may be true or false, while inferences are the intermediate processes that connect premises to conclusions by selecting, relating, and evaluating relevant information [27].

To carry out practical reasoning activities, domain knowledge is also necessary in addition to logic principles [199]. For example, mathematical knowledge is necessary for arithmetic reasoning, whereas commonsense knowledge is necessary for reasoning in everyday tasks. In addition to the provided inputs, the domain knowledge acts as extra premises and is essential to drawing reliable conclusions in particular disciplines.

Humans infer conclusions from texts that help them make sense of the story being told. In order for machines to possess this skill, they need to be able to learn relevant and accurate commonsense for an infinite number of scenarios. Although commonsense reasoning is a natural and effortless skill for humans, artificial intelligence systems have historically struggled to replicate it, and current models exhibit partial and task-dependent forms of commonsense reasoning. They often struggle to make intuitive decisions about everyday circumstances relying on patterns learned from data rather than exhibiting genuine understanding of social and physical contexts [113]. This is partly because most artificial intelligence systems are trained on datasets and objectives specific to a given task, which produces models that are good at identifying correlations specific to a given task but devoid of clear-cut, understandable commonsense reasoning [36][88][112].

In recent years, LLMs have performed relatively well on reasoning-based tasks [29]. The degree to which these models can reason similarly to humans has been the subject of discussion [118]. Skeptics contend that the performance of the models primarily reflects their ability to memorize the vast amount of data they have been trained on [190], while proponents argue that reasoning capabilities emerge with scale, referring to LLMs as foundation models [22]. The main obstacle in finding out whether LLMs have true reasoning abilities is due to the enormous size, complexity, and closed-source nature of popular models and the training data they use. Furthermore, in-depth examinations of the reasoning behavior of the models are sometimes overshadowed by the focus on how well they perform on downstream reasoning tasks [56].

Commonsense reasoning in LLMs relies heavily on memorization and the appropriate application of information pertaining to individuals, locations, and events that happen in our everyday lives. Due to their extensive real-world expertise, LLMs excel at knowledge-intensive activities. For example, they are able to perform well on a variety of general question-answering benchmarks. However, when faced with tasks that just require contextual knowledge or when the knowledge requirements do not align with their prior knowledge, LLMs do not perform as well as fine-tuned models [198]. Investigating the relationship between memorization and reasoning would provide information on LLMs, promoting a deeper comprehension and offering suggestions for improving these skills.

Tasks for evaluating knowledge memorization and application (like reasoning) are included in several benchmarks to support the research of memory-reasoning correlates. However, because they employed the distinct and already-existing datasets for various tasks, there were no inherent links between these activities. Despite the profusion of applications and approaches, a systematic review on the application of commonsense reasoning in LLMs is lacking. Such a review would organize and clarify the distinctions and relationships between these works. This paper seeks to give an in-depth review of the current state of commonsense reasoning in LLMs, illuminating their potential, applications, and future directions.

The main contributions of this survey are summarized as follows:

- This research provides a comprehensive survey of commonsense reasoning in LLMs, covering key datasets, models, benchmarks, and enhancement techniques, while detailing both the advancements and limitations in this area. It serves as a valuable resource for understanding the current capabilities and challenges of LLMs in emulating human-like reasoning.
- This research explores enhancement techniques and research opportunities aimed at improving commonsense reasoning in LLMs, and outlines potential directions for addressing current challenges in the field.
- This research identifies and discusses critical issues that LLMs face in commonsense reasoning, such as such as hallucinations, biases, and limitations in current datasets—emphasizing the obstacles these models encounter in replicating human-like reasoning. These insights underscore the areas where current systems struggle to achieve nuanced commonsense understanding.
- This research emphasizes the need for ongoing efforts to enhance LLMs' commonsense reasoning capabilities. It suggests incorporating mechanisms such as external knowledge bases or reasoning modules to improve the reliability of these models, highlighting future directions for research and development in this field.

The remainder of this paper is organized as follows: Section 2 reviews recent work. Section 3 describes the methodology used in this survey, including the time period covered and the inclusion and exclusion criteria for the literature. Section 4 surveys and analyzes existing work, including commonsense reasoning datasets, LLM models for commonsense reasoning, and quantitative results on commonsense reasoning benchmarks across different LLMs. Section 5 surveys and discusses methods for enhancing commonsense reasoning in LLMs. Section 6 analyzes and discusses potential opportunities and challenges related to commonsense reasoning in LLMs. Finally, Section 7 concludes the paper and suggests directions for future research.

2 Recent Works

Several recent surveys highlight the progress, methodologies, and challenges involved in equipping LLMs with commonsense capabilities [30, 82, 86, 95, 133, 141]. These works reflect growing interest in how LLMs can mimic or approximate human-like understanding in everyday reasoning tasks, ranging from causal inference to social norm interpretation.

A notable survey by Richardson and Heck reviews the integration of commonsense knowledge into LLMs used for conversational AI [141]. The study demonstrated that although transformer-based models, such as BERT and GPT-3, have advanced significantly, they frequently have trouble answering queries that call for implicit world knowledge. This survey divided methods into two main categories: prompting or fine-tuning models with curated reasoning datasets, and incorporating organized commonsense knowledge (e.g., ConceptNet [153], ATOMIC [149]) into models. The authors stressed that while programs such as BlenderBot and LaMDA demonstrate fluency at the surface level, they fall short in situations that call for more complex commonsense reasoning.

In the assessment of explainable AI and commonsense reasoning in language models, Krause and Stolzenburg [86] paid special attention to LLMs such as GPT-3.5 [203], Gemma [166], and LLaMA 3 [173]. According to their investigation, these models performed better than humans on multiple-choice commonsense tests (e.g., classification or prediction tasks [69]), but they have trouble with activities that call for logical multi-step reasoning. More importantly, they investigate how well model outputs match human explanations, discovering that even in cases where their ultimate responses are accurate, many LLMs offer erroneous or insufficient explanations.

Another recent work is the comprehensive taxonomy of prompt-based commonsense reasoning by Plaat et al. [133], which categorizes prompting strategies for multi-step reasoning. The survey

examines explanation-based prompting, self-consistency sampling, and CoT prompting. Additionally, it detects enduring problems like hallucinated reasoning processes and prompt sensitivity. Instead of focusing only on the end response, the authors advocate for improved evaluation criteria that account for the quality and correctness of intermediate reasoning processes.

The absence of multilingual commonsense benchmarks is addressed by Lin et al. [95], who point out that the majority of commonsense evaluations concentrate on datasets that are only available in English. Evaluation across many linguistic and cultural contexts is made possible by their development of the Mickey corpus, a multilingual collection of more than half a million commonplace statements in 11 languages. Researchers can evaluate how well LLMs generalise commonsense reasoning outside of English using the Mickey Probe, a probing task created in tandem with the corpus that highlights significant barriers to cross-lingual transfer. Alfugaha and AL-Smadi [4] recently conducted a focused study on SemEval-2020 Task 4, which involves commonsense validation and explanation. Their survey examines how well modern LLMs, such as LLaMA 3-70B, perform in recognising implausible claims and choosing plausible answers. They found that large models have great detection accuracy, but their explanations are frequently illogical or excessively general, highlighting the disconnect between classification performance and actual reasoning abilities.

In addition to performance-focused surveys, some recent works have investigated the internal mechanisms behind commonsense reasoning. Using layer attribution and attention visualisation, Joshi et al. [82] investigate how LLMs handle reasoning tasks across layers and tokens, determining when and how commonsense appears during inference. This study provides insights into interpretability and model debugging by identifying particular latent pathways in models such as GPT-3.5 that become active in response to reasoning-heavy prompts. Moreover, complementing these surveys are meta-analyses such as those by Chang et al. [30], which benchmark over 30 different commonsense reasoning datasets and models, comparing not only accuracy but also robustness, fairness, and generalization. When presented with hostile or out-of-distribution cases, they reveal a consistent disparity in reasoning performance, suggesting that current benchmarks might overstate actual capabilities.

Although commonsense reasoning in LLMs has been the subject of several previous assessments, our work stands out in three important aspects. First, we offer a comprehensive and methodical survey that covers datasets, model architectures, commonsense reasoning benchmarks, as well as a quantitative analysis of different commonsense reasoning datasets—unlike previous surveys, which often focused narrowly on specific model families or datasets. Second, by including a thorough meta-analysis of benchmark data and emphasizing subtle differences among model tiers, we go beyond superficial performance summaries. Third, our study incorporates emerging enhancement methods—such as Chain-of-Thought (CoT) prompting, logical fallacy detection, and hybrid symbolic-neural reasoning frameworks—into a coherent research trajectory. By taking this comprehensive approach, we not only describe prior research but also provide practical guidance on addressing persistent challenges of commonsense reasoning in LLMs.

3 Method

This survey includes studies conducted between January 2020 and March 2025 on commonsense reasoning in LLMs, with additional discussion of significant developments through early 2026—particularly the emergence of reasoning-focused models that employ test-time compute scaling. The time period was selected to encompass current developments in LLMs, especially those that make use of knowledge integration strategies, transformer architectures, and large-scale pretraining. GPT-3, available in 2020 with 175 billion parameters, set new records in LLM capabilities. Its large scale allowed it to perform a wide range of tasks with minimal fine-tuning. GPT-3 was one of the

first models to demonstrate how powerful LLMs could be when trained on massive amounts of data. Therefore, it is crucial to review research from this era. The focus on commonsense reasoning is key, as it demonstrates how LLMs are not just learning syntactic patterns from data but are also acquiring knowledge that aligns more closely with human-like understanding. Previous studies, like [35] and [25], point to recent strides in improving LLMs' ability to handle commonsense knowledge, which is essential for tasks requiring deeper contextual awareness or nuanced decision-making.

We included studies that specifically address commonsense reasoning in LLMs to provide a thorough yet targeted review. The importance of commonsense reasoning in natural language comprehension has been demonstrated by seminal works like [149] and [161]. Papers that offer empirical assessments, fresh benchmark datasets (such CommonsenseQA [163], OpenbookQA [115], PIQA [21], and SocialIQA [150]), and creative training techniques that improve LLMs' reasoning abilities are all included in this survey. We took into account studies that were published in high-impact AI and NLP journals as well as peer-reviewed conferences including NeurIPS, ACL, EMNLP, AAAI, ICML, and ICLR. To promote producibility and future research, preference was given to studies that offer open-access datasets, code, or model implementations.

Research that has not been empirically evaluated—such as theoretical debates without experimental validation—was not included in our analysis. Although knowledge graphs and symbolic AI support commonsense reasoning [201], research that only examined non-LLM techniques was excluded. Additionally disqualified were workshop articles that did not offer significant contributions, short abstracts, and extended abstracts. Furthermore, publications written in languages other than English were excluded in order to preserve uniformity and accessibility. Finally, in order to ensure that our analysis concentrates on the most current and pertinent developments, we eliminated studies that were redundant or out-of-date and had been replaced by more recent research.

Google Scholar, Semantic Scholar, and scholarly databases including IEEE Xplore, arXiv, and ACL Anthology were used to perform a thorough literature search. We utilized phrases such as "commonsense reasoning in LLMs," "large language models and commonsense knowledge," "transformers and commonsense reasoning," with "benchmarking LLM commonsense." To guarantee thorough coverage of significant publications, the citation networks of important papers, including those by [98] and [188], were also investigated. Following an initial screening process based on keywords and abstracts, full-text papers were examined to make sure they met the inclusion requirements.

4 Existing Work on Commonsense Reasoning in LLMs

4.1 Datasets

Research has been spurred in numerous domains of commonsense, supported by the creation of diverse datasets as shown in Table 1. Table 1 provides an overview of various commonsense reasoning datasets released between 2018 and 2025 and commonly used in publications from 2020 to 2025. Each dataset is listed along with its release year, a brief description of what it entails, and the specific focus it targets. These datasets enable evaluating different commonsense capability of LLMs [3, 17, 21, 28, 42, 52, 57, 60, 99, 101, 117, 136, 147, 162, 163, 185, 202].

On many datasets, current systems have accuracies comparable to human performance (typically around 80%) [28, 136]. Most researchers find it challenging to replicate the success of state-of-the-art models because these models rely on extensive pre-trained corpora that are often difficult to access and require significant computational resources [17]. Achieving similar performance demands adherence to numerous criteria, including fine-tuning strategies, hyperparameter optimization, and architectural refinements, which can be complex and resource-intensive [42]. Furthermore, in addition to leveraging large pre-trained models, successful approaches often necessitate further

Dataset	Year	Description	Focus
SWAG [207]	2018	Predicts plausible event continuations from video descriptions	event sequence prediction
AI2 ARC [33]	2018	Complex scientific and commonsense QA	Deep reasoning and problem-solving
ARC-C [33]	2018	Challenge partition of ARC for more difficult questions	Challenging scientific QA
OpenbookQA [115]	2018	QA requiring facts from a science book and commonsense reasoning	Open-book reasoning
SocialIQA [150]	2019	Questions about social interactions and dynamics	Social intelligence reasoning
HellaSwag [208]	2019	Challenging questions on narrative event sequences	Advanced event reasoning
QASC [83]	2020	QA dataset requiring multi-sentence reasoning using science facts	Multi-hop science QA
PIQA [21]	2020	Tests physical commonsense about object usage	Physical reasoning
Numersense [94]	2020	Evaluates numerical commonsense in language models	Numerical reasoning
ProtoQA [23]	2020	Open-ended QA with graded human judgment	Graded plausibility and answer variety
RICA Probes [216]	2020	Human-verified probes for zero-shot reasoning	Zero-shot narrative reasoning
WinoGrande [147]	2021	Expanded Winograd Schema Challenge [91] focusing on pronoun resolution	Ambiguity resolution
ParaCOMET [57]	2021	Generates paragraph-level commonsense inferences using structured knowledge (e.g., ATOMIC)	Generative commonsense inference
HyKAS [107]	2021	Performs zero-shot commonsense reasoning by integrating symbolic and neural knowledge sources	Hybrid zero-shot commonsense reasoning
StrategyQA [162]	2021	Questions that require multi-hop reasoning over implicit facts	Implicit and multi-hop commonsense reasoning
MC-TACO [117]	2021	Temporal commonsense questions with multiple correct answers	Temporal reasoning
CommonsenseQA 2.0 [163]	2022	Enhanced version of CommonsenseQA with tougher questions	Complex reasoning
Vera [99]	2022	Scores commonsense plausibility in knowledge graphs	Plausibility scoring
Gemini Reasoning Dataset [185]	2023	Multimodal tasks combining text and vision	Multimodal reasoning
ConvoSense [53]	2024	Probes dialog-based commonsense understanding through natural conversations	Conversational commonsense reasoning
mCSQA [148]	2024	Multilingual CommonsenseQA extending CommonsenseQA to multiple languages	Multilingual commonsense QA
COM2 [49]	2024	Logical queries over commonsense graphs	Multi-hop reasoning

Table 1. Evaluation Datasets for Commonsense Reasoning in LLMs, Released between 2018 and 2025

training on task-specific datasets and external knowledge sources such as knowledge graphs. This additional training process adds another layer of complexity, making it even more difficult for researchers to achieve comparable results without substantial resources and expertise [52, 202].

The same running and tuning procedure should be repeated multiple times when applying them to different tasks in order to determine the optimal fit. In order to use commonsense reasoning, LLMs must not only retain factual knowledge, but also make inferences regarding the facts. Research has shown that LLMs continue to outperform fine-tuned models on the majority of datasets, including

ARC-C [33] and StrategyQA [60]. In particular, GPT-4 has performed nearly 100% (96.3%) [3] on ARC-C, which is a set of challenging questions on science tests for grades 3 through 9.

To assess the models' comprehension of real language and their capacity to reason using external commonsense information, a number of difficult WSC-like (Winograd Schema Challenge [91]) problems and question-answer tasks have been created. Additionally, ComonsenseQA [163], was built based on ConceptNet [153]. It is intended as a five-choice QA dataset that needs a model to capture the relationship between the question and the right response.

More recently, Liu et al. utilised conceptual dependence theory to create PrimeNet, a new commonsense knowledge base. Their research showed that, because of the numerous intuitive functions created, PrimeNet has high-quality commonsense knowledge that may be applied to commonsense reasoning in a variety of downstream tasks [101]. Overall, a number of datasets, including HellaSwag [208], SocialIQA [150], PIQA [21], CommonsenseQA 2.0 [163], Vera [99], and COM2 [49], have been introduced with the goal of encouraging commonsense reasoning. Davis provided a few recommendations for creating new datasets to probe commonsense reasoning [35]. It is especially important to ensure the datasets are high quality, to prioritize fundamental commonsense information over encyclopedic, expert, and common knowledge, and to broaden the application of current benchmarks in order to address reasoning with higher structural complexity and richness.

Table 2 provides an overview of pretraining and fine-tuning datasets for commonsense reasoning in LLMs. This table categorizes datasets based on their role in pretraining, fine-tuning, dataset splits, and multimodal reasoning. The datasets cover a broad range of commonsense knowledge, including general world knowledge, social reasoning, causal and temporal reasoning, counterfactual reasoning, and ethical considerations.

Category	Datasets and Description
Pretraining Datasets for Commonsense Reasoning	
Web and Books Corpora	C4 (Colossal Clean Crawled Corpus) [43], The Pile [58]
Social Media and Dialogues	OpenWebText [55], Reddit (Pushshift.io) [15], Social media datasets [145]
Knowledge Graphs	ConceptNet [153], ATOMIC [149], WordNet [51], WikiData [178], DBPedia [12]
Fine-Tuning Datasets for Commonsense Reasoning	
Commonsense QA and Reasoning	CommonsenseQA [163], Social IQa [150], PIQA [21], HellaSwag [208], ARC-C [33], SWAG [207], Winograd Schema Challenge [91]
Causal and Temporal Reasoning	COPA (Choice of Plausible Alternatives) [143], TIMEDIAL [134], Event2Mind [137]
Ethical and Social Commonsense	ETHICS [39], Moral Stories [47], ToxiGen [64]
Multimodal Commonsense Reasoning Datasets	
Vision-Language Tasks	VisualCommonsenseQA (VCR) [215], VisualCOMET [130], Winoground [171], NLVR2 [157]
Dataset Splits for Commonsense Reasoning	
Training Set	Typically 70–80% of the dataset for model training [20]
Validation Set	Typically 10–15% for hyperparameter tuning [20]
Test Set	Typically 10–15% for final evaluation [20]
Adversarial/OOD Splits	Explicitly used in HellaSwag and SWAG via adversarial filtering to evaluate generalization [207, 208]

Table 2. Overview of Pretraining, Fine-Tuning, and Multimodal Datasets Used for Commonsense Reasoning, along with Their Descriptions and Dataset Split Criteria.

The development of commonsense reasoning in LLMs usually involves combining several pretraining and fine-tuning datasets. While social media sites like Reddit and OpenWebText improve conversational commonsense, pretraining datasets include extensive corpora like C4, The Pile, Wikipedia, and Common Crawl that offer extensive exposure to general knowledge [108]. ConceptNet [153], ATOMIC [149], and WikiData [178] structured knowledge aids models in comprehending causal and inferential linkages [66]. CommonsenseQA [161], Social IQa [150], PIQA [21], HellaSwag [208], and other fine-tuning datasets measure situational and physical reasoning, whereas COPA [143] and Event2Mind [137] evaluate causal and temporal understanding [137].

To make sure models understand societal norms and biases, datasets such as Moral Stories and Counterfactual QA concentrate on hypothetical and ethical reasoning [129]. Usually, these datasets are divided into test, validation, and training sets; some even include adversarial splits to evaluate the generalization of the model. Datasets such as VisualCommonsenseQA [215] and Winoground [171] assess text-image comprehension for multimodal commonsense reasoning [171]. New developments use multi-hop reasoning, synthetic data synthesis, and few-shot learning to boost performance [96].

Dataset selection is essential for improving model capabilities in certain domains as LLMs develop because commonsense reasoning is more motivated by in-context learning rather than explicit fine-tuning.

4.2 Models

Commonsense reasoning datasets—such as CommonsenseQA [163], OpenbookQA [115], PIQA [21], and SocialIQA [150], as discussed in the previous section—have been developed to train or evaluate the performance of language models such as GPT-4 [3], Qwen [195], Phi [1], Claude [194], and Gemini [165]. This section presents a comprehensive overview of prominent language models employed in commonsense reasoning tasks.

Table 3 summarizes a wide range of models, highlighting their key features and specializations relevant to tasks involving implicit knowledge, and everyday reasoning. The models can broadly be categorized into three groups: General-purpose pretrained models, Task-specific fine-tuned models, and Knowledge-augmented or logic-based models.

General-purpose models such as GPT-3/4 [85], OPT [211], Codex [111], and Claude [194] have shown strong performance on commonsense benchmarks when adapted through few-shot learning or instruction tuning. Lightweight variants like ALBERT [193] and Phi [1] emphasize compactness while maintaining competitive reasoning capabilities.

Task-specific models like T5 [151], BART [104], and UnifiedQA [158] adopt a text-to-text framework and are fine-tuned on benchmark datasets such as SWAG, PIQA, and CommonsenseQA. These models are designed to generalize across multiple commonsense reasoning formats.

Knowledge-augmented models integrate structured data to enhance reasoning. ERNIE [159] and K-BERT [102] incorporate external knowledge graphs (e.g., ConceptNet), while NLM [100] combines neural networks with symbolic logic for rule-based learning. These models aim to explicitly embed real-world facts and relationships to improve inference.

Recent advancements have introduced powerful open-source and instruction-tuned models such as LLaMA [173], Falcon [5], Vicuna [32], Gemma [166], and Qwen [195]. These models vary in scale and capability, with some supporting multimodal inputs and multilingual reasoning.

Finally, cutting-edge models like Deepseek [97], Gemini [165], and Claude [194] represent the state-of-the-art in safe, generalizable, and multimodal commonsense AI systems. While some, like Codex, were originally developed for code generation, they have also proven adaptable for reasoning tasks through careful prompt design.

Model	Description
GPT-3/ GPT-4 [85]	Large models with general knowledge. Fine-tuned for commonsense tasks like Winograd Schema and COPA.
T5 [151]	Frames all tasks as text-to-text problems. Fine-tuned on datasets like SWAG and CommonsenseQA for reasoning.
UnifiedQA [158]	Fine-tuned on a variety of question-answering datasets, including CommonsenseQA.
BART [104]	A sequence-to-sequence model pre-trained for text recovery. Fine-tuned for tasks like PIQA and CommonsenseQA.
ALBERT [193]	A lightweight variant of BERT, tuned for commonsense tasks like CommonsenseQA.
ERNIE [159]	Integrates commonsense knowledge from graphs like ConceptNet, improving reasoning.
K-BERT [102]	Adds knowledge graph info into BERT for better commonsense reasoning.
NLM [100]	Combines neural networks with logic-based reasoning for rule learning.
OpenAI Codex [111]	Primarily for code, but can be adapted for commonsense reasoning with structured prompts.
LLaMA [173]	A collection of foundation language models ranging from 7B to 65B parameters.
Falcon [5]	Causal decoder-only models trained on a diverse high-quality corpora predominantly assembled from web data.
OPT [211]	Trained to roughly match the performance and sizes of the GPT-3 class of models.
Gemma [166]	A family of cutting-edge, lightweight open models developed using the same technology and research as Gemini models.
Vicuna [32]	An open-source chatbot trained by fine-tuning LLaMA on user-shared conversations collected from ShareGPT.
Qwen [195]	A family of Transformer-based large and multimodal models by Alibaba Cloud, supporting text, code, vision, and audio, fine-tuned with SFT/RLHF for assistant-style performance across languages and modalities.
Phi [1]	A compact "textbook-quality" LLM series by Microsoft, achieving SOTA reasoning and programming performance in its size class, under MIT license and without RLHF.
Mistral [76]	Known for achieving high-quality results with fewer parameters.
Deepseek [97]	A strong Mixture-of-Experts language model with 671B total parameters with 37B activated for each token.
Claude [194]	A family of LLMs by Anthropic designed with a focus on constitutional AI for steerable, safe, and helpful behavior.
Gemini [165]	A family of multimodal models by Google DeepMind, designed for reasoning across text, image, and code, building on PaLM and incorporating innovations from AlphaCode and Flamingo.
Reasoning-Focused Models	
OpenAI o1/o3 [127]	Reasoning-focused models that employ test-time compute scaling through extended chain-of-thought reasoning, allocating additional inference-time computation to solve complex problems step-by-step.
DeepSeek-R1 [38]	A reasoning model trained via pure reinforcement learning without human-annotated reasoning demonstrations, enabling emergent reasoning strategies including self-reflection and verification.
QwQ [135]	An open-source reasoning model by Alibaba with 32B parameters, employing extended thinking to match larger reasoning models on mathematical and logical tasks.

Table 3. Overview of Language Models for Commonsense Reasoning and Their Descriptions.

A significant recent development is the emergence of *reasoning-focused models* that employ test-time compute scaling. Unlike standard LLMs, which generate responses in a single forward pass, models such as OpenAI's o1 [127] (released September 2024), o3 (released April 2025), and DeepSeek-R1 [38] (released January 2025) allocate additional computational resources at inference time to produce explicit, step-by-step reasoning traces before arriving at a final answer. DeepSeek-R1 is particularly notable for demonstrating that reasoning capabilities can emerge through pure reinforcement learning without human-annotated chain-of-thought demonstrations. Open-source alternatives such as QwQ-32B [135] by Alibaba (released March 2025) have brought similar reasoning capabilities to smaller, accessible model sizes. These reasoning-focused models have achieved remarkable results on mathematical and coding benchmarks—for example, o3 achieves 96.7% on AIME and DeepSeek-R1 achieves 97.3% on MATH-500—but their impact on commonsense reasoning is more nuanced, as discussed in Section 5.

Overall, the models listed in Table 3 reflect the evolution of commonsense reasoning in language models—from transformer-based architectures and unified task formulations to integration of structured knowledge , multimodal capabilities, and most recently, inference-time reasoning strategies. This diversity illustrates the range of strategies being used to build AI systems.

4.3 Quantitative Results on Commonsense Reasoning Benchmarks

To further analyze the commonsense reasoning capabilities of LLMs, this section presents a comprehensive survey of quantitative results on reasoning benchmarks across major datasets, followed by a meta-analysis that further analyzes these results to uncover valuable discoveries about commonsense reasoning in LLMs.

4.3.1 ***Quantitative Benchmark Results from GPT-3 to ChatGPT (2023)*** . The evaluation of commonsense reasoning capabilities in large language models gained particular momentum following the release of ChatGPT in November 2022. One of the comprehensive investigations was conducted by Bian et al. (2024), whose work initially appeared on arXiv in March 2023 [18], providing timely insights into the commonsense abilities of ChatGPT when it was powered by the GPT-3.5 model. This independent research established important baselines that align with the official metrics later reported in OpenAI's GPT-4 Technical Report [125].

Dataset	Domain	GPT-3	Instruct GPT	ChatGPT	Human
CommonsenseQA	General	38.0	**81.0**	74.0	88.9
OpenBookQA	General	22.0	65.0	**73.0**	89.3
PIQA	Physical	48.0	77.0	**78.0**	94.5
Social IQA	Social	36.0	**71.0**	62.0	86.9
MC-TACO	Temporal	20.0	**53.0**	52.0	75.8
HellaSWAG	Event	19.0	61.0	**67.0**	95.7
NumerSense	Numerical	45.0	63.0	**79.0**	89.7
QASC	Science	25.0	**75.0**	74.0	93.0
ARC	Science	27.0	88.0	**94.0**	–
ProtoQA	Prototypical	67.3	84.6	**94.2**	–

Table 4. Early evaluation results (% accuracy) of LLMs on commonsense reasoning tasks demonstrate the progression from GPT-3 to ChatGPT (GPT-3.5) across diverse commonsense reasoning datasets [18].

The evaluation results presented in Table 4 reveal several important patterns that established the foundation for understanding LLM capabilities in commonsense reasoning [18].

The progression from GPT-3 to Instruct GPT and ChatGPT demonstrates the substantial impact of instruction tuning and human alignment techniques. Across all domains, both Instruct GPT and ChatGPT achieved dramatically higher accuracies compared to the base GPT-3 model, with improvements often exceeding 30-40%.

Even in these early evaluations, distinct patterns emerged across different commonsense domains. ChatGPT showed particularly strong performance on science-related tasks (ARC: 94.0%, ProtoQA: 94.2%) and numerical reasoning (NumerSense: 79.0%), while struggling more with social (Social IQA: 62.0%) and temporal reasoning (MC-TACO: 52.0%). This pattern of domain-specific performance gaps has persisted in subsequent model generations.

Despite the improvements shown by ChatGPT over earlier models, substantial gaps remained between AI and human performance across most domains, with differences ranging from 5-30%. These early findings highlighted that while instruction tuning significantly improved commonsense reasoning capabilities, fundamental challenges remained in achieving human-level understanding.

These findings by Bian et al. [18] provided crucial insights that informed the development of more sophisticated evaluation frameworks and highlighted the specific areas where future models would need improvement. The patterns established in this foundational work continue to influence our understanding of commonsense reasoning capabilities in current state-of-the-art models.

4.3.2 ***Recent Benchmark Performance with Quantitative Results.*** Recent results demonstrate that state-of-the-art LLMs have achieved remarkable performance on major commonsense reasoning benchmark datasets [8, 18, 89, 125, 208], with leading models approaching or even surpassing human baselines on specific tasks as shown in Table 5. Table 5 presents a performance comparison of state-of-the-art LLMs on commonsense reasoning benchmarks, with most results published between 2023 and 2025.

The top-performing models demonstrate convergence toward human parity on HellaSwag [208], with Claude 3 Opus achieving 95.4% accuracy and GPT-4 reaching 95.3%, both approaching the 95.6% human baseline. This near-saturation performance suggests that HellaSwag may be approaching the limits of its discriminative power for evaluating incremental improvements in future models, consistent with broader concerns about benchmark obsolescence in rapidly advancing AI systems [26].

More substantial challenges were discovered for LLMs working with the WinoGrande [147] dataset, revealing a consistent 5.5–6.5% performance deficit even among the most advanced models as shown in Table 5. Claude 3 Opus (88.5%) and GPT-4 (87.5%) both fall meaningfully short of the 94.0% human baseline, indicating that contextual ambiguity resolution and pronoun reference tasks continue to challenge current architectures. This gap suggests that certain fundamental aspects of commonsense reasoning—particularly those requiring sophisticated contextual understanding—remain incompletely solved.

As shown in Table 5, ARC-C [33] demonstrates the most complex performance landscape, with leading models like Claude 3 Opus (96.4%) and GPT-4 (96.3%) achieving very high scores that potentially exceed human performance, though direct human baseline comparisons remain unavailable. However, the substantial variation across model families—with Llama models scoring significantly lower (71-74%)—suggests that performance on knowledge-intensive reasoning tasks may be particularly sensitive to training methodology and data composition.

CommonsenseQA [163] shows how well various language models perform on a multiple-choice benchmark designed to test commonsense reasoning, using a zero-shot setting (no in-context examples). Human performance is 88.9%, and top models like GPT-4o and Qwen 2 72B come close at 87.3%. Other strong performers include Claude 3.5 Sonnet (84.3%), Llama 3.1 70B (84.4%), and Phi-3 Small (80.3%), showing that both proprietary and open models can achieve high accuracy.

In contrast, earlier or smaller models like Llama 2 7B (54.6%) and Mistral 7B (68.0%) perform significantly worse.

4.3.3 ***Meta-Analysis: A Three-Tier Framework of LLM Reasoning Capabilities.*** Our meta-analysis of benchmark results reveals a three-tier hierarchy in current LLM reasoning performance:

Tier 1: Near-Human Performance (GPT-4, Claude 3 Opus). These models achieve 95%+ on commonsense benchmarks like HellaSwag and strong results on contextual reasoning (e.g., 87.5–88.5% on WinoGrande) as shown in Table 5. Their strength lies in associative reasoning and pattern recognition within familiar contexts, driven by extensive training data and advanced architectures. However, they still struggle with causal reasoning and novel scenarios.

Tier 2: High but Inconsistent Performance (Claude 3 Sonnet, Llama 3.x series). As shown in Table 5, models in this tier score well on certain tasks (e.g., 93.2% on ARC-C for Claude 3 Sonnet) but show significant drops in others (e.g., 75.1% on WinoGrande). Llama models perform well on pattern-based tasks (84–88%) but lag in abstract reasoning (71–74% on ARC-C). Their capabilities are strong but domain-specific.

Tier 3: Persistent Reasoning Limits (All Models). Across all models, systematic reasoning gaps remain—especially in tasks like WinoGrande, where there's still a 5.5–6.5% gap from human performance as shown in Table 5. These reflect the current limits of LLM reasoning, regardless of architecture.

As shown in Table 5, some models released in 2024–2025, such as GPT-4.1, Claude 4 Opus, Gemini 2.5 Pro, and Llama 4 have no reported results yet on some major datasets. These findings reveal gaps that need to be filled.

The quantitative evidence synthesized in this analysis establishes a critical foundation for understanding both the remarkable empirical achievements and fundamental theoretical limitations of current LLMs in commonsense reasoning. While these models have demonstrated unprecedented progress toward human-level performance on specific evaluation benchmarks, the underlying cognitive mechanisms appear to rely predominantly on sophisticated statistical pattern recognition rather than the systematic logical reasoning that characterizes human commonsense understanding. These findings highlight the significant gap that remains between current AI systems and truly human-like commonsense reasoning capabilities, underscoring the need for continued research on more fundamental approaches to machine reasoning with commonsense.

5 Enhancing Commonsense Reasoning in LLMs

Recent research highlights that prompting techniques like chain-of-thought (CoT) can enhance LLMs' reasoning abilities [84, 160, 183, 187]. Thinking routines may further support reasoning in LLMs by structuring prompts to guide deeper thought [205], thereby enhancing their commonsense reasoning. However, LLMs still struggle with tasks such as detecting logical fallacies—skills that are critical for evaluating argument quality and combating misinformation across political, social, and media contexts—underscoring the need for continued improvements in reasoning accuracy [2, 13, 93, 146]. Thus, improving the ability to detect logical fallacies may contribute to more valid and robust commonsense reasoning in LLMs. This section surveys and discusses various strategies aimed at enhancing commonsense reasoning in LLMs.

Model	HellaSwag	WinoGrande	ARC-C	CommonsenseQA
Human Performance	95.6-95.7%[18, 208]	94.0%[147]	94.0%[33]	88.9-90%[18, 40, 192]
GPT 4	95.3%$^{(10\text{-}shot)}$[125]	87.5%$^{(5\text{-}shot)}$[125]	96.3%$^{(25\text{-}shot)}$[125]	83.2%$^{(0\text{-}shot)}$[40]
GPT 3.5	85.5%$^{(10\text{-}shot)}$[125]	81.6%$^{(5\text{-}shot)}$[125]	85.2%$^{(25\text{-}shot)}$[125]	85.2%$^{(25\text{-}shot)}$[125]
Claude 3 Opus	95.4%$^{(10\text{-}shot)}$[8]	88.5%$^{(5\text{-}shot)}$[8]	96.4%$^{(25\text{-}shot)}$[8]	88%$^{(0\text{-}shot)}$[168]
Claude 3 Sonnet	89.0%$^{(10\text{-}shot)}$[8]	75.1%$^{(5\text{-}shot)}$[8]	93.2%$^{(25\text{-}shot)}$[8]	82%$^{(0\text{-}shot)}$[168]
Llama 3.1 405B	88.4%$^{(10\text{-}shot)}$[89]	85.0%$^{(5\text{-}shot)}$[89]	73.7%$^{(25\text{-}shot)}$[89]	89%$^{(0\text{-}shot)}$[168]
Llama 3.3 70B	86.4%$^{(10\text{-}shot)}$[89]	84.0%$^{(5\text{-}shot)}$[89]	71.7%$^{(25\text{-}shot)}$[89]	86%$^{(0\text{-}shot)}$[168]
GPT 4o	∼95%$^{(est.)}$[126]	86.5%$^{(0\text{-}shot)}$[154]	95.3%$^{(0\text{-}shot)}$[154]	87.3%$^{(0\text{-}shot)}$[154]
GPT 4o mini	91.0%$^{(0\text{-}shot)}$[189]	71.9%$^{(0\text{-}shot)}$[154]	82.6%$^{(0\text{-}shot)}$[154]	83.9%$^{(0\text{-}shot)}$[154]
GPT 4.1	95.8%$^{(10\text{-}shot)}$[164]	–	–	–
Claude 3 Haiku	85.9%$^{(10\text{-}shot)}$[7]	66.2%$^{(0\text{-}shot)}$[154]	89.3%$^{(0\text{-}shot)}$[154]	79.0%$^{(0\text{-}shot)}$[154]
Claude 3.5 Sonnet	89.0%$^{(10\text{-}shot)}$[10]	85.7%$^{(0\text{-}shot)}$[154]	95.3%$^{(0\text{-}shot)}$[154]	84.3%$^{(0\text{-}shot)}$[154]
Claude 3.5 Haiku	93.0%$^{(10\text{-}shot)}$[9]	62.0%$^{(0\text{-}shot)}$[131]	94.0%$^{(25\text{-}shot)}$[11]	83%$^{(0\text{-}shot)}$[168]
Claude 4 Opus	95.4%$^{(10\text{-}shot)}$[75]	–	–	–
Claude 4 Sonnet	72.7%$^{(0\text{-}shot)}$[123]	90.3%$^{(0\text{-}shot)}$[19]	–	–
Gemini 1.5 Flash	86.5%$^{(10\text{-}shot)}$[41]	74.8%$^{(0\text{-}shot)}$[154]	93.6%$^{(0\text{-}shot)}$[154]	82.6%$^{(0\text{-}shot)}$[154]
Gemini 1.5 Pro	93.3%$^{(10\text{-}shot)}$[59]	78.3%$^{(0\text{-}shot)}$[154]	90.6%$^{(0\text{-}shot)}$[154]	82.9%$^{(0\text{-}shot)}$[154]
Gemini 2.5 Pro	–	–	–	87%$^{(0\text{-}shot)}$[213]
Gemini 2.5 Flash	–	–	–	85%$^{(0\text{-}shot)}$[213]
Llama 2 7b	77.2%$^{(0\text{-}shot)}$[174]	50.4%$^{(0\text{-}shot)}$[154]	45.2%$^{(0\text{-}shot)}$[154]	54.6%$^{(0\text{-}shot)}$[154]
Llama 3.1 8b	82.33%$^{(10\text{-}shot)}$[124]	63.3%$^{(0\text{-}shot)}$[154]	82.6%$^{(0\text{-}shot)}$[154]	74.9%$^{(0\text{-}shot)}$[154]
Llama 3.1 70b	86.7%$^{(10\text{-}shot)}$[140]	81.2%$^{(0\text{-}shot)}$[154]	93.6%$^{(0\text{-}shot)}$[154]	84.4%$^{(0\text{-}shot)}$[154]
Llama 4	–	78.9%$^{(5\text{-}shot)}$[139]	–	–
DeepSeek R1	85.0%$^{(0\text{-}shot)}$[37]	85.0%$^{(5\text{-}shot)}$[139]	92.2%$^{(0\text{-}shot)}$[170]	86.3%$^{(0\text{-}shot)}$[103]
Qwen 2 7b	68.3%$^{(10\text{-}shot)}$[197]	60.7%$^{(0\text{-}shot)}$[154]	75.3%$^{(0\text{-}shot)}$[154]	79.0%$^{(0\text{-}shot)}$[154]
Qwen 2 72b	87.6%$^{(10\text{-}shot)}$[71]	80.7%$^{(0\text{-}shot)}$[154]	94.6%$^{(0\text{-}shot)}$[154]	87.3%$^{(0\text{-}shot)}$[154]
Qwen 3	83.0%$^{(0\text{-}shot)}$[214]	90.0%$^{(0\text{-}shot)}$[195]	65.3%$^{(0\text{-}shot)}$[196]	87.5%$^{(0\text{-}shot)}$[195]
Mistral 7b	82.8%$^{(10\text{-}shot)}$[138]	56.5%$^{(0\text{-}shot)}$[154]	76.6%$^{(0\text{-}shot)}$[154]	68.0%$^{(0\text{-}shot)}$[154]
Gemma 2 9b	81.9%$^{(10\text{-}shot)}$[167]	67.7%$^{(0\text{-}shot)}$[154]	89.6%$^{(0\text{-}shot)}$[154]	80.1%$^{(0\text{-}shot)}$[154]
Phi 3 Small 8k	76.7%$^{(10\text{-}shot)}$[6]	81.6%$^{(0\text{-}shot)}$[154]	91.0%$^{(0\text{-}shot)}$[154]	80.3%$^{(0\text{-}shot)}$[154]

Table 5. Performance comparison of state-of-the-art LLMs on major commonsense reasoning datasets. All scores represent accuracy (%), and most of the results were published between 2023 and 2025. Where specified in the source, the number of in-context examples (shots) used for evaluation is noted. Data has been compiled from official technical reports, public leaderboards, and publications. Missing values indicate scores not publicly available.

5.1 Chain-of-Thought Prompting

LLMs can be directed to create "reasoning" expressly in order to encourage them to reason instead of just giving responses. According to recent research, LLMs can behave in ways that are similar to "reasoning" when given specific prompts. Chain-of-thought (CoT) prompting, for example asks models to explicitly list the processes they took to arriving to a solution [187]. This strategy entails giving LLMs a few instances of "chain of thought", or intermediary natural language reasoning processes, in the prompt. Specifically, ⟨input, output⟩ demonstrations are replaced with ⟨input, chain of thought, output⟩ triples. In this way, the model learns to produce explicit justification before coming up with the final response, given a target question [187].

Experimental findings demonstrate that this straightforward concept can sometimes significantly enhance the few-shot performance on arithmetic, symbolic, and commonsense reasoning tasks in LLMs. CoT is shown to enhance language model performance on multi-step reasoning, problem-solving, and Q&A tasks in English and multilingual contexts in both the few-shot and zero-shot prompting styles [84][160][187]. Moreover, it has been demonstrated that CoT functions in multimodal contexts [144][212].

The core idea behind CoT prompting is that by showing models examples of reasoning pathways, they can better approximate human-like problem-solving. Wei et al. showed that prompting the PaLM 540B model with as few as eight CoT exemplars achieved state-of-the-art accuracy on the GSM8K benchmark of math word problems, surpassing even finetuned GPT-3 with verification modules [187]. Building on this, Wang et al. introduced the self-consistency decoding method, which samples multiple reasoning paths instead of relying on a single greedy chain [184]. This technique further boosts CoT prompting performance across several benchmarks such as GSM8K (+17.9%), SVAMP (+11.0%), AQuA (+12.2%), StrategyQA (+6.4%), and ARC-challenge (+3.9%) [183].

Evaluating the faithfulness, robustness, and utility of CoT explanations is an active area of research [78][74]. For instance, Wei et al. systematically assessed CoT reasoning on commonsense reasoning tasks, highlighting the importance of interpretability and reliability in real-world applications. Meanwhile, Turpin et al. showed that CoT can be manipulated to generate deceptively plausible yet incorrect justifications, underscoring the need for careful scrutiny [175].

In parallel, efforts to integrate external knowledge—via knowledge graphs, retrieval augmentation, or fine-tuning with structured data—seek to enhance the factual consistency and reasoning accuracy of CoT outputs [52][200][128]. Such approaches may complement CoT prompting by grounding reasoning steps in verified information sources, thus improving both performance and trustworthiness [202][210][50].

Despite the notable empirical success of the concept and the emergence of a theoretical understanding, there is still much to learn about the mechanical underpinnings of the success of CoT [110][182]. CoT transcripts superficially mimic human reasoning [44][187] and enhance the capability of commonsense reasoning in LLMs, even though it is unclear whether the observed output truly reflects the internal reasoning processes of the language model. In the next subsection, we will provide a detailed quantitative analysis of CoT effectiveness, leveraging the latest research.

5.2 Quantitative Analysis of Chain-of-Thought (CoT) Effectiveness

While CoT prompting is often highlighted as a method for enhancing reasoning in LLMs, empirical evidence reveals a more complex picture. To provide a comprehensive analysis of CoT effectiveness, we synthesize and reanalyze the experimental results from [154], presenting novel aggregations and cross-model comparisons that reveal patterns not explicitly highlighted in the original work. Our analysis across multiple models and five commonsense reasoning datasets shows that CoT's effectiveness is highly dependent on the model, the task, and whether few-shot examples are provided.

5.2.1 ***An Aggregated Quantitative Analysis.*** Table 6 presents our aggregated analysis of average performance across all models for each dataset, comparing zero-shot and few-shot prompting strategies where applicable. This cross-dataset synthesis, derived from the individual results reported in [154] (complete original results and methodology are documented in Appendix A), reveals notable patterns that emerge from this comprehensive view: while most datasets show mixed results with CoT prompting in zero-shot settings, StrategyQA demonstrates substantial improvement (+4.3% with zero-shot and +5.4% with few-shot CoT prompting), and WinoGrande shows consistent positive gains (+2.3% improvement).

Performance variations become more pronounced when examined by individual model, as demonstrated in our model-centric analysis in Table 7. This cross-dataset aggregation by model, synthesized from [154], reveals significant disparities in model capabilities, with GPT-4o achieving the highest zero-shot performance and Llama 3.1 70b leading in few-shot scenarios.

	Zero-Shot Avg (%)			Few-Shot Avg (%)		
Dataset	**Direct**	**CoT**	**Diff**	**Direct**	**CoT**	**Diff**
CommonsenseQA	**79.2**	77.9	−1.3	74.1	**74.8**	0.8
StrategyQA	68.7	**73.0**	4.3	70.3	**75.7**	5.4
PIQA	86.4	**86.4**	0.1	–	–	–
SocialIQA	**68.2**	66.9	−1.3	–	–	–
WinoGrande	71.8	**74.1**	2.3	–	–	–
Overall Average	74.8	**75.7**	0.9	72.2	**75.3**	3.1

Table 6. Performance Comparison of Direct and CoT Prompting across Major Commonsense Reasoning Datasets: The values represent our calculated averages across all evaluated models based on results from [154]. The highest performance for each dataset-setting combination is highlighted in bold. Overall averages are calculated across all datasets for zero-shot setting, and only across datasets with available few-shot data for the few-shot settings. "–" indicates that the results or data are not available in [154].

	Zero-Shot Avg (%)			Few-Shot Avg (%)		
Model	**Direct**	**CoT**	**Diff**	**Direct**	**CoT**	**Diff**
Claude-3 Haiku	74.3	77.2	2.9	–	–	–
Claude-3.5 Sonnet	84.3	85.8	1.5	–	–	–
GPT-4o Mini	81.8	83.2	1.4	**85.4**	83.8	−1.6
Gemini 1.5 Flash	80.3	76.8	−3.5	81.8	79.4	−2.4
Gemini 1.5 Pro	80.4	78.3	−2.1	–	–	–
Gemma 2 9b	75.0	76.1	1.1	77.0	79.4	2.4
GPT-4o	**87.3**	**87.7**	0.4	–	–	–
Meta-Llama 2 7b	51.4	50.9	−0.5	25.0	38.0	**13.0**
Meta-Llama 3.1 70b	84.2	84.7	0.5	84.4	**85.9**	1.5
Meta-Llama 3.1 8b	72.9	73.4	0.5	71.2	74.2	3.0
Mistral 7b	58.3	61.8	3.5	71.2	72.2	1.0
Phi-3 Small 8k	70.8	72.5	1.7	75.8	77.9	2.2
Qwen 2 72b	82.9	84.9	2.0	83.9	85.8	1.8
Qwen 2 7b	64.0	66.1	2.1	65.9	76.1	10.2
Average	74.8	75.7	0.9	72.2	75.3	3.1

Table 7. Performance Comparison of Direct and CoT Prompting across Models and Settings: Performance is shown for both zero-shot and few-shot settings with direct and CoT promptingm and the numbers are based on the results from [154]. Bold values indicate the highest performance in each column. Notable findings include Meta-Llama 2 7b's remarkable 13.0% improvement with few-shot CoT and Qwen 2 7b's substantial 10.2% gain with few-shot CoT. Average rows show our calculated overall performance across all models.

5.2.2 ***Zero-Shot Performance***. In zero-shot settings, where models receive no examples, our analysis of the [154] results shows that CoT prompting modestly outperforms direct answering across the evaluated commonsense tasks.

The effect of zero-shot CoT varies by dataset:

- **Overall Trend**: On average across the five datasets, using CoT results in a **+0.9%** improvement in accuracy compared to direct answering as shown in Table 6.

- **Dataset-Specific Results**: The positive impact of CoT is most pronounced on **StrategyQA** (+4.3% improvement) and **WinoGrande** (+2.3% improvement) as shown in Table 6. **PIQA** shows minimal improvement (+0.1%), while **CommonsenseQA** (−1.3%) and **SocialIQA** (−1.3%) show slight negative impact.

The effect of zero-shot CoT also varies significantly by model:

- **Models Benefiting from CoT**: Several models show substantial zero-shot advantages as shown in Table 7. The most consistent performers are **Mistral 7b** (+3.5%), **Claude-3 Haiku** (+2.9%), and **Qwen 2 7b** (+2.1%), followed by **Qwen 2 72b** (+2.0%) and **Phi-3 Small 8k** (+1.7%).
- **Models Hindered by CoT**: Some models perform worse with zero-shot CoT. The largest performance drops are seen in **Gemini 1.5 Flash** (−3.5%), **Gemini 1.5 Pro** (−2.1%), and **Meta-Llama 2 7b** (−0.5%) as shown in Table 7.

5.2.3 ***Few-Shot Performance.*** The dynamics change in few-shot contexts, where providing examples makes CoT an even more effective strategy.

- **Overall Trend**: As shown in Table 6, for datasets with few-shot evaluations, CoT provides a clear advantage with an average improvement of **+3.1%**. On **StrategyQA**, CoT maintains strong improvement (+5.4%), and on **CommonsenseQA**, the trend reverses from negative to a **+0.8%** gain.
- **Model-Specific Turnaround**: The models that struggled most with zero-shot CoT see the largest gains from few-shot examples. **Meta-Llama 2 7b** shows a dramatic improvement (+13.0%) and **Qwen 2 7b** achieves substantial gains (+10.2%) as shown in Table 7.
- **High-Performing Model Behavior**: Interestingly, as shown in Table 7, models that performed well in zero-shot settings, like **GPT-4o Mini** and **Gemini 1.5 Flash**, show slight performance degradation (−1.6% and −2.4% respectively) when using CoT in few-shot settings.

5.2.4 ***Implications and Recommendations.*** Our reanalysis of the [154] findings reveals a more nuanced picture than the assumption that CoT is universally beneficial and leads to more refined recommendations:

- **Task-Dependent Effectiveness**: CoT shows clear benefits on tasks requiring multi-step reasoning (like StrategyQA and WinoGrande) but may hinder performance on tasks that rely on direct factual recall or simple pattern matching (like CommonsenseQA and SocialIQA).
- **Model and Task Sensitivity**: The vast difference in performance underscores that the effectiveness of a prompting strategy is a function of the model's architecture and the specific task. The **Meta-Llama 2 7b** model, for instance, is a poor zero-shot reasoner but becomes dramatically improved with few-shot CoT.

5.2.5 ***Summary: Choosing Between Direct and CoT Prompting.***

- **Use CoT for Multi-Step Reasoning**: For tasks requiring structured reasoning (like StrategyQA) or complex inference (like WinoGrande), CoT prompting provides consistent benefits across both zero-shot and few-shot settings.
- **Be Cautious with Simple Tasks**: For straightforward commonsense tasks that may rely on direct factual knowledge, direct prompting often remains competitive or superior, especially in zero-shot settings.
- **Leverage Few-Shot CoT for Weaker Models**: When working with smaller or less capable models, providing a few examples of CoT reasoning can unlock significant performance gains, particularly for complex reasoning tasks.

- **Test Both Strategies**: Given the high variability across models and tasks, practitioners should evaluate both direct and CoT prompting. Models like **Claude-3 Haiku** and **Mistral 7b** excel with CoT across settings, while models like **GPT-4o Mini** show mixed responses depending on the prompting context.

5.3 Detecting Logical Fallacies to Ensure Valid Reasoning

LLMs still struggle with more difficult tasks such as accurately identifying logical fallacies. Drawn from philosophy, logical fallacies are errors in reasoning [172]. They typically occur when the premises are insufficient or irrelevant to reach the conclusions. In argumentative discourse, finding fallacies is essential to assessing the quality of the argument for valid reasoning [179][122]. Furthermore, identifying fallacies is especially helpful for critical thinking tools and disinformation detection systems toward valid reasoning.

Aristotle studied fallacies from antiquity and described them as an argumentative technique used to mislead an opponent and obtain the upper hand in a dispute [54]. With time, this definition changed [176][62], bringing the concept of the Aristotelian fallacy into line with more contemporary conceptions of mathematics and logic. Fallacies are defined as "important, baptizable types of errors or deceptive tactics of argumentation that tend to fool or mislead participants in everyday discussions" [181]. This definition, proposed by Walton, Reed, and Macagno [181], reflects a more modern and comprehensive understanding of fallacies compared to earlier, purely logical interpretations, by emphasizing their role in real-world argumentation and discourse. This term is more appropriate for the problems with natural language that we might encounter nowadays and is less restrictive. Fallacies can be found in various domains [169], such as political arguments [13, 61], and social media [109]. They are also a component of the propaganda strategies used to advance particular agendas and influence public opinion. Fallacies were particularly prominent during the 2020 COVID-19 vaccine controversy [46], when fake news circulated on news sites and social media [13, 146].

Previous research [80][152] have demonstrated that LLMs had difficulty in identifying logical fallacies, with F1 scores hovering around 66% and only reaching 27% following transfer learning. However, Li et al. showed that GPT-4 can recognize and classify fallacies, obtaining over 86% accuracy in both identification and classification tasks [93]. Prior studies have also looked at the the standpoints of deductive reasoning, natural language inference, and reading comprehension [98][63]. However, there is a lack of research on logical fallacies, which is in fact the major reason causing logical inconsistency in LLMs. In recent years, studies on logical fallacies have mostly concentrated on fallacy categorization [61][80][132] and dataset construction [114][80]. For example, Jin et al. initially presented a framework of the 13 types of logical fallacies and assessed all sentence samples on a classification task when they suggested the goal of logical fallacy detection.

In addition, developments in deep learning and the availability of more data have led to the development of methods for automatically identifying and categorizing fallacies in text [13][146][2]. Despite this, the study is disjointed, with most methods concentrating on particular corpora (like speeches) or fallacies (such as ad hominem fallacies exclusively).

Furthermore, there is disagreement over a common vocabulary, not all works employ the same kinds of fallacies, and there are differing degrees of specificity when it comes to fallacies. Generally, identifying fallacies is a subjective endeavor by nature. Although earlier works have acknowledged this subjectivity, none have openly embraced it.

5.4 Reasoning-Focused Models and Test-Time Compute Scaling

A major development since mid-2024 has been the rise of *reasoning-focused models* that employ test-time compute scaling to enhance reasoning capabilities [38, 127, 135]. Unlike standard LLMs

that generate responses in a single forward pass, these models—including OpenAI's o1 and o3 series, DeepSeek-R1, and QwQ—allocate additional computational resources at inference time to produce explicit, multi-step reasoning traces before arriving at a final answer. This paradigm represents a fundamentally different approach to enhancing reasoning compared to techniques such as CoT prompting: whereas CoT prompting elicits reasoning through carefully crafted instructions, reasoning-focused models are trained (often via reinforcement learning) to generate extended internal deliberation autonomously.

These models have demonstrated impressive gains on mathematical and coding benchmarks. For example, DeepSeek-R1 achieves 97.3% on MATH-500 and 71.0% on AIME 2024 [38], while OpenAI's o3 achieves 96.7% on AIME [127]. However, their impact on commonsense reasoning tasks is more nuanced. Emerging evidence suggests that extended reasoning chains can actually *harm* performance on commonsense tasks—a phenomenon termed "overthinking" [31]. When additional computation is applied to tasks that humans solve intuitively, models may over-analyze the problem, becoming distracted by irrelevant considerations and producing incorrect answers. This finding is consistent with the observation that commonsense reasoning often relies on rapid, intuitive pattern matching rather than explicit multi-step logical deduction.

Furthermore, recent evaluations using compositional reasoning benchmarks such as Logical-CommonsenseQA [14] have revealed that reasoning-focused models, while performing well on conjunctive reasoning (combining multiple positive constraints), show sharp performance degradation on negation-based compositions. This suggests that the extended thinking paradigm does not uniformly address the diverse reasoning patterns required for commonsense understanding.

These findings carry important implications for the field. First, they highlight that the nature of commonsense reasoning—which is largely implicit, fast, and grounded in everyday experience—may be fundamentally different from the mathematical and logical reasoning tasks where test-time compute scaling excels. Second, they suggest that future approaches to enhancing commonsense reasoning should focus not on longer reasoning chains but on improving inference-time knowledge retrieval and compositional reasoning capabilities. Third, the adaptive reasoning approaches emerging in recent models—which dynamically adjust reasoning depth based on task complexity—represent a promising middle ground, potentially avoiding the overthinking problem while still benefiting from extended deliberation on genuinely complex commonsense tasks.

5.5 Future Directions for Enhancing Commonsense Reasoning in LLMs

Despite their outstanding performance in many NLP tasks, LLMs still face a substantial problem when it comes to commonsense reasoning. Besides the enhancements mentioned in the above subsections—such as CoT prompting, leveraging thinking routines, detecting logical fallacies, and reasoning-focused models —this subsection will highlight other future directions for enhancing commonsense reasoning in LLMs.

Although LLMs are capable of producing fluid and contextually appropriate writing, they often struggle with tasks that humans consider simple, such as understanding physical laws, social conventions, or cause-and-effect relationships [86]. To address this, future research could focus on incorporating structured knowledge, enhancing training techniques, and developing more robust evaluation frameworks. Another promising direction is the improved integration of explicit knowledge sources into LLMs. Currently, models rely heavily on implicit knowledge learned during pretraining, which can be incomplete or biased. By integrating structured commonsense knowledge bases like ConceptNet [153], ATOMIC [149], or Cyc [90]—through fine-tuning or retrieval-augmented generation (RAG)—LLMs could base their reasoning on validated facts. Additionally, rather than being constrained by static training data, models could dynamically update their

knowledge to incorporate the most recent information, enabling them to reason with up-to-date data.

The creation of hybrid symbolic-neural techniques is another important field. Although LLMs are excellent at identifying patterns, they frequently lack the rigorous logical reasoning needed for challenging everyday tasks [87]. Consistency and dependability may be increased by combining neural models with symbolic reasoning engines, such as logic-based systems or theorem provers. An LLM might, for example, provide logical steps that are believable and then checked for logical consistency by a symbolic system. This collaboration may result in reasoning that is stronger and easier to understand. Since surface-level pattern matching can be used to game many of the current benchmarks, evaluation is still a crucial challenge. Implicit reasoning, adversarial robustness, and practicality should be prioritised in future benchmarks. Deeper understanding of how well models function in real-world situations may be possible through human-in-the-loop assessments. Furthermore, as demonstrated by models such as GPT-4V [3] and Gemini [73], multimodal commonsense reasoning—which integrates visual, hearing, and sensor data—could assist models in gaining a more grounded grasp of the physical world.

Lastly, long-term advancement of LLMs may be fueled by human-AI cooperation and self-improvement techniques. The reasoning depth of LLMs could be improved by methods like recursive decomposition of difficult issues into simpler sub-tasks or iterative self-correction, in which models evaluate and improve their own reasoning. Addressing defects in reasoning such as biases and knowledge gaps in LLMs may also be facilitated by crowdsourcing human input on model faults [209]. As the field evolves, the goal is to move beyond fragile, black-box models and develop LLMs that can explain their decisions, adapt to new situations, and communicate naturally with humans.

6 Potential Opportunities and Challenges

The integration of commonsense reasoning into LLMs opens up a wide range of transformative opportunities. One major opportunity lies in enhancing human-AI collaboration, where models can better understand human goals, intentions, and context, enabling smoother and more intuitive interactions. In creative fields such as content generation, storytelling, and game design, commonsense reasoning allows LLMs to produce more coherent and believable narratives that align with real-world logic and social norms. In the legal and policy domains, it can assist in interpreting complex regulations and anticipating real-world implications of certain decisions. For robotics and embodied AI, commonsense reasoning can enable more intelligent decision-making in dynamic environments, allowing machines to act with greater autonomy and reliability. Additionally, in the development of personalized learning and mental health support systems, commonsense-aware LLMs can tailor responses that are emotionally intelligent and context-sensitive. As research advances, the potential to create AI systems that reason more like humans promises not only to expand the utility of LLMs but also to increase trust and safety in AI applications.

There are other exciting potentials when commonsense reasoning is incorporated into LLMs. Commonsense reasoning in LLMs significantly enhances their real-world applicability across a wide range of domains. In customer service, LLMs equipped with commonsense understanding can interpret vague or implicit user queries and provide more relevant, human-like responses. In healthcare, they can assist with symptom checking and patient triage by inferring logical connections between symptoms and potential conditions. Educational tools benefit from LLMs' ability to offer context-aware explanations, adapt to learners' needs, and identify misunderstandings. In robotics and autonomous systems, commonsense reasoning enables LLMs to interpret and respond appropriately to dynamic environments and human intentions. Moreover, in content moderation, LLMs can better detect subtle forms of harmful or misleading content by understanding

social context and intent. These use cases demonstrate how integrating commonsense reasoning allows LLMs to perform more reliably, ethically, and effectively in complex, real-world scenarios.

Despite significant advancements, LLMs continue to face persistent challenges that hinder their ability to perform robust commonsense reasoning. One major issue is hallucination [24], where models generate plausible-sounding but factually incorrect or misleading information. This undermines trust in their outputs, especially in critical applications. Additionally, LLMs often inherit and amplify societal biases [17] present in the data they are trained on, leading to unfair or prejudiced responses. Dataset limitations [120] further compound these issues—many existing benchmarks lack the diversity, contextual depth, or real-world complexity needed to fully evaluate or train models on nuanced reasoning tasks. Together, these challenges highlight the importance of developing more reliable evaluation frameworks, curating higher-quality datasets, and incorporating mechanisms to detect and mitigate hallucinations and bias in future iterations of LLMs.

The emergence of reasoning-focused models (Section 5.4) also presents both opportunities and challenges for commonsense reasoning. On one hand, the ability to allocate variable compute at inference time opens new possibilities for tackling complex commonsense problems that require multi-step deliberation—such as reasoning about causal chains, anticipating consequences of actions, or resolving ambiguous social situations. On the other hand, the "overthinking" phenomenon [31] suggests that commonsense reasoning may be fundamentally different from mathematical or logical reasoning in its computational requirements, and that simply scaling inference-time computation is insufficient. This distinction underscores the need for approaches specifically tailored to commonsense reasoning, rather than assuming that general-purpose reasoning enhancements will transfer to everyday understanding.

LLMs that integrate structured knowledge sources [156] are emerging as powerful tools for enhancing commonsense reasoning capabilities. By combining the generative strengths of LLMs with structured knowledge bases such as ConceptNet [153], ATOMIC [149], and CSKG [72] during inference, these models can improve commonsense reasoning and reason more effectively. This integration allows the models to compensate for the limitations of purely statistical language generation, especially in scenarios requiring real-world reasoning or domain-specific knowledge [106]. Furthermore, emphasizing explainability [45] enables these hybrid models to provide transparent, human-interpretable justifications for their responses, helping users better understand and trust the outputs by elucidating the models' reasoning paths.

7 Conclusion

In conclusion, LLMs have made notable strides toward commonsense reasoning, but significant challenges remain. These models show promise in emulating human-like reasoning but often fall short in capturing the intricate inferences required for a deep understanding of both physical and social contexts. Issues such as inadequate contextual information, limitations in comparative reasoning, and domain-specific knowledge gaps hinder their ability to fully replicate human commonsense reasoning. Notably, the recent emergence of reasoning-focused models—such as OpenAI's o1/o3 series and DeepSeek-R1—which employ test-time compute scaling and extended chain-of-thought reasoning, has yielded dramatic improvements on mathematical and coding benchmarks but has not comparably advanced commonsense reasoning performance. The "overthinking" phenomenon observed in these models suggests that commonsense reasoning may require fundamentally different approaches than the explicit multi-step deliberation that benefits formal reasoning tasks. Furthermore, the reliance on LLMs raises concerns about the potential impact on human creativity, critical thinking, and decision-making, given their propensity to generate misleading or inaccurate information. Despite these hurdles, the progress achieved underscores the potential of LLMs to improve reasoning capabilities. To address the existing limitations and enhance model reliability,

ongoing research and development are essential. This includes integrating robust mechanisms for fact-checking, developing adaptive reasoning strategies that match computational effort to task requirements, and continuous optimization to better harness the potential of LLMs. As the field evolves, addressing these challenges will be crucial for advancing the effectiveness and accuracy of commonsense reasoning in artificial intelligence.

Commonsense reasoning is vital for many natural language processing activities. It determines causation and evaluates the relative plausibility of various possibilities. Up until now, formalizing commonsense reasoning knowledge and mathematical logic have been the main subjects of research. Replicating commonsense reasoning in artificial intelligence is a significant challenge since it requires machines to comprehend and interpret the vast and complex knowledge humans take for granted. Formal rules or algorithms cannot easily capture the inferences made by commonsense reasoning about the physical and social world. Furthermore, some LLMs continue to face problems, such as inadequate contextual information, issues with comparative reasoning, and gaps in expertise in specific domains like science and medicine.

The reliance on artificial intelligence programs could also have detrimental effects on human qualities such as creativity, critical thinking, and academic integrity. Another important area of concern is that LLMs have the potential to produce inaccurate or misleading information that could impact decision-making. Therefore, future models could benefit from incorporating fact-checking mechanisms to mitigate these issues. Overall, while LLMs show promise and are progressing continuously, they still face significant obstacles and limitations, necessitating further thorough investigation and optimization.

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A Detailed Chain-of-Thought Performance Analysis

This appendix presents a comprehensive empirical analysis comparing the performance of Chain-of-Thought (CoT) prompting with direct answering across multiple large language models and commonsense reasoning datasets. The results and methodology are based on data reported by [154].

A.1 Methodology

Datasets: Five widely-used commonsense reasoning benchmarks representing different aspects of everyday reasoning were used. Table 8 provides an overview of these datasets, including the type of reasoning, answer format, and the number of examples provided in few-shot prompts.

Dataset	Reasoning Type	Answer Format	m-Shots
CommonsenseQA	Commonsense	Multiple choice	7
StrategyQA	Commonsense	True or False	6
PIQA	Commonsense	Multiple choice	0
SocialIQA	Commonsense	Multiple choice	0
WinoGrande	Commonsense	Multiple choice	0

Table 8. Summary of Datasets Used in Experiments

Models: Fourteen representative LLMs spanning different architectures, sizes, and training methodologies, including both open-source models (Llama, Mistral, Qwen, Phi-3, Gemma) and proprietary models (GPT-4, Claude, Gemini).

Evaluation Protocol: Standardized zero-shot and few-shot prompting with consistent formatting for both direct and CoT conditions. Direct prompting used a simple question-answer format, while CoT prompting included explicit instructions to show reasoning steps.

Metrics: Accuracy scores calculated as the percentage of correct answers, averaged across all test instances for each dataset-model combination.

A.2 Comprehensive Performance Results

A.2.1 Zero-Shot Performance Analysis. Table 9 presents the complete performance comparison across all evaluated models and datasets for the zero-shot scenario. The table shows accuracy percentages for both direct answering and Chain-of-Thought prompting, along with the difference (CoT - Direct).

Dataset	Model	Direct (%)	CoT (%)	Difference
	Llama 2 7b	54.6	49.4	−5.2
	Mistral 7b	68.0	68.0	0.0
	Llama 3.1 8b	74.9	68.5	−6.4
	Llama 3.1 70b	84.4	83.5	−0.9
	Gemma 2 9b	80.1	79.2	−0.9
	Phi-3 Small 8k	80.3	81.8	1.5
CommonsenseQA	Qwen 2 7b	79.0	78.5	−0.5

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Table 9 continued from previous page

Dataset	Model	Direct (%)	CoT (%)	Difference
	Qwen 2 72b	87.3	87.4	0.1
	GPT-4o Mini	83.9	82.5	−1.4
	GPT-4o	87.3	86.5	−0.8
	Claude-3 Haiku	79.0	80.6	1.6
	Claude-3.5 Sonnet	84.3	85.1	0.8
	Gemini 1.5 Flash	82.6	79.7	−2.9
	Gemini 1.5 Pro	82.9	79.9	−3.0
PIQA	Llama 2 7b	64.7	62.1	−2.6
	Mistral 7b	77.7	78.6	0.9
	Llama 3.1 8b	84.2	85.0	0.8
	Llama 3.1 70b	90.6	91.8	1.2
	Gemma 2 9b	84.8	84.0	−0.8
	Phi-3 Small 8k	85.5	89.1	3.6
	Qwen 2 7b	86.2	84.3	−1.9
	Qwen 2 72b	89.1	92.9	3.8
	GPT-4o Mini	88.6	93.1	4.5
	GPT-4o	95.5	95.9	0.4
	Claude-3 Haiku	86.6	85.9	−0.7
	Claude-3.5 Sonnet	94.5	94.6	0.1
	Gemini 1.5 Flash	89.8	84.6	−5.2
	Gemini 1.5 Pro	91.3	88.1	−3.2
SocialIQA	Llama 2 7b	55.9	53.4	−2.5
	Mistral 7b	33.5	35.9	2.4
	Llama 3.1 8b	73.5	73.5	0.0
	Llama 3.1 70b	80.9	78.7	−2.2
	Gemma 2 9b	76.3	74.9	−1.4
	Phi-3 Small 8k	40.4	38.0	−2.4
	Qwen 2 7b	39.3	37.3	−2.0
	Qwen 2 72b	80.4	80.5	0.1
	GPT-4o Mini	80.0	79.0	−1.0
	GPT-4o	81.5	81.9	0.4
	Claude-3 Haiku	74.8	75.4	0.6
	Claude-3.5 Sonnet	81.0	79.7	−1.3
	Gemini 1.5 Flash	79.1	74.5	−4.6
	Gemini 1.5 Pro	78.2	73.9	−4.3
StrategyQA	Llama 2 7b	31.2	39.5	8.3
	Mistral 7b	55.8	66.1	10.3
	Llama 3.1 8b	68.6	73.7	5.1
	Llama 3.1 70b	83.8	85.3	1.5
	Gemma 2 9b	66.4	73.7	7.3
	Phi-3 Small 8k	66.0	72.3	6.3
	Qwen 2 7b	54.8	63.2	8.4
	Qwen 2 72b	76.9	81.7	4.8

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Table 9 continued from previous page

Dataset	Model	Direct (%)	CoT (%)	Difference
	GPT-4o Mini	84.5	82.2	−2.3
	GPT-4o	85.5	84.5	−1.0
	Claude-3 Haiku	65.0	73.4	8.4
	Claude-3.5 Sonnet	76.3	80.1	3.8
	Gemini 1.5 Flash	75.2	72.5	−2.7
	Gemini 1.5 Pro	71.4	74.0	2.6
WinoGrande	Llama 2 7b	50.4	49.9	−0.5
	Mistral 7b	56.5	60.4	3.9
	Llama 3.1 8b	63.3	66.5	3.2
	Llama 3.1 70b	81.2	84.2	3.0
	Gemma 2 9b	67.7	68.7	1.0
	Phi-3 Small 8k	81.6	81.5	−0.1
	Qwen 2 7b	60.7	67.1	6.4
	Qwen 2 72b	80.7	81.9	1.2
	GPT-4o Mini	71.9	79.2	7.3
	GPT-4o	86.5	89.7	3.2
	Claude-3 Haiku	66.2	70.7	4.5
	Claude-3.5 Sonnet	85.7	89.4	3.7
	Gemini 1.5 Flash	74.8	72.5	−2.3
	Gemini 1.5 Pro	78.3	75.5	−2.8

Table 9. Complete Zero-Shot Performance Results: Direct Answering vs Chain-of-Thought Reasoning

A.2.2 Few-Shot Performance Analysis. In addition to zero-shot evaluation, this section presents a performance analysis based on few-shot prompting, where the model is provided with a small number of examples (7 for CommonsenseQA and 6 for StrategyQA) before being asked to answer. This approach tests the model's ability to learn from context. Note that unlike the zero-shot evaluation which includes all 14 models, the few-shot analysis reports results for only 10 models. The results are shown in Table 10.

Dataset	Model	Direct (%)	CoT (%)	Difference
CommonsenseQA (7-shot)	Llama 2 7b	19.2	18.2	−1.0
	Mistral 7b	70.4	73.6	3.2
	Llama 3.1 8b	76.5	74.0	−2.5
	Llama 3.1 70b	84.6	84.7	0.1
	Gemma 2 9b	80.8	81.8	1.0
	Phi-3 Small 8k	80.4	80.8	0.4
	Qwen 2 7b	72.9	80.3	7.4
	Qwen 2 72b	87.8	88.4	0.6
	GPT-4o Mini	84.7	84.7	0.0
	Gemini 1.5 Flash	83.3	81.7	−1.6

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Table 10 continued from previous page

Dataset	Model	Direct (%)	CoT (%)	Difference
StrategyQA (6-shot)	Llama 2 7b	30.9	57.9	27.0
	Mistral 7b	72.0	70.7	−1.3
	Llama 3.1 8b	65.8	74.4	8.6
	Llama 3.1 70b	84.2	87.1	2.9
	Gemma 2 9b	73.3	77.1	3.8
	Phi-3 Small 8k	71.1	75.0	3.9
	Qwen 2 7b	58.9	71.9	13.0
	Qwen 2 72b	80.1	83.2	3.1
	GPT-4o Mini	86.2	83.0	−3.2
	Gemini 1.5 Flash	80.3	77.0	−3.3

Table 10. Few-Shot Performance: Direct Answering vs. Chain-of-Thought Reasoning

B Reasoning Types

Research using LLMs to address logical reasoning tasks, including commonsense reasoning [147], has recently increased due to the rapid improvement of LLMs. Reasoning in LLMs encompasses various types that reflect how these models process and generate information to solve problems or answer questions [34, 147, 155]. Commonsense reasoning—the ability to make sound judgments about everyday situations using general world knowledge—is not a standalone form of reasoning, but rather draws upon and interacts with different types of reasoning. Each type of reasoning serves a specific purpose and is applied in different contexts to make sense of information and draw conclusions. Table 11 compares different types of reasoning philosophies, outlining their distinct features, typical examples, applications, and their relationship with commonsense reasoning.

Type of Reasoning	Characteristics	Example	Applications	Relationship with Commonsense Reasoning
Deductive [48]	$\Rightarrow$ General to specific; $\therefore$ conclusion must be true if premises are true	Humans are mortal. Socrates is human. $\Rightarrow$ Socrates is mortal.	Mathematics, Logic	Commonsense reasoning can leverage deductive steps in familiar contexts (e.g., "All fruits have seeds; apples are fruits; so apples have seeds").
Inductive [65]	$\Rightarrow$ Specific to general; $\approx$ conclusion is probable	The sun rises daily. $\Rightarrow$ The sun will rise tomorrow.	Science, Empirical Studies	Inductive reasoning is central to commonsense, as both involve generalizing from experience—for example, inferring it might rain when people are seen carrying umbrellas on a cloudy day.
Abductive [206]	$\Rightarrow$ Observe a phenomenon; infer most likely explanation	Doctor observes symptoms. $\Rightarrow$ Infers likely condition.	Diagnostics, Hypothesis Generation	Commonsense often uses abductive reasoning to explain everyday situations. For example, if you hear the doorbell and see a delivery truck, you infer a package arrived.
Analogical [186]	$\equiv$ Reasoning by similarity	Earth has water. Exoplanet X has water. $\Rightarrow$ Exoplanet X may support life.	Modeling, Problem Solving	Analogical reasoning supports commonsense reasoning by allowing us to understand new situations through similarities with familiar experiences.
Causal [79]	$\rightarrow$ Cause and effect relationship	Ice melted. $\rightarrow$ Temperature rose.	Science, History	Causality is fundamental to commonsense understanding. Knowing that pushing a glass causes it to fall is a causal insight rooted in commonsense.
Practical [180]	$\rightsquigarrow$ Action-oriented reasoning	Want health $\rightsquigarrow$ Eat well, exercise	Decision Making	Commonsense often deals with deciding what to do (e.g., crossing the street safely), making practical reasoning a key operational layer.
Moral [191]	$\Rightarrow$ Ethical evaluation	Harm is wrong $\Rightarrow$ Factory farming is wrong	Ethics, Policy	Many commonsense decisions involve basic moral judgments—like sharing food or helping someone in distress.
Modal [81]	$\diamond$ Possibility; $\Box$ Necessity	$\diamond$ Wallabies might win season opener; $\Box$ Vehicles must stop at red lights	Metaphysics, Logic	Modal reasoning is a core enabler of commonsense flexibility, allowing it to deal with future planning, and counterfactual scenarios
Probabilistic [119]	$\sim$ Involves uncertainty or likelihood	80% of people who hear a sound also see a flash $\Rightarrow$ If you hear the sound, likely a flash follows	Risk, Statistics	Commonsense reasoning often operates under uncertainty, using informal heuristics or "gut feelings" to make judgments (e.g., "It's cloudy—probably going to rain").

Table 11. Comparison of different types of reasoning philosophies, outlining their characteristics, typical examples, applications, and their relationship with commonsense reasoning.

Deductive reasoning ensures logically valid conclusions from true premises but is limited by its inability to generate new knowledge or handle ambiguity [48].

Inductive reasoning involves drawing generalizations or predictions from known information and underpins many everyday cognitive processes such as decision-making, classification, and scientific inference [65].

Abductive reasoning starts with an observation or set of observations and seeks the most plausible explanation for them. It's often described as inference to the best explanation [206].

Analogical reasoning - thinking based on analogy - is fundamental to human cognition and widely used in AI tasks such as machine translation, computer vision, and ranking. In NLP, it is often modeled through vector embeddings like Word2vec [116], which use vector offsets to capture word analogies, though this method has known limitations [92].

Causal reasoning consists in inferring and using causal relations. Decision making, problem solving, and learning often require causal reasoning. It is essential for decision-making, learning, and advancing artificial general intelligence by enabling LLMs to move beyond correlations toward deeper explanatory and predictive capabilities [67].

Practical reasoning is the process of determining what action to take, guided by principles of rationality and influenced by moral, prudential, and value-based considerations. It distinguishes itself from theoretical reasoning, which focuses on what to believe [177].

Moral reasoning involves making judgments about what individuals should do in social contexts, often studied through hypothetical scenarios, though real-world complexities—such as those observed on social media—offer deeper insights into everyday moral decision-making [77].

Modal reasoning involves considering what is possible, necessary, or contingent, enabling individuals to evaluate alternative scenarios and outcomes, which is essential for critical thinking and widely applied in fields like philosophy, linguistics, and computer science [16].

Probabilistic reasoning provides a mathematical framework for managing uncertainty, enabling AI systems to make informed decisions in real-world scenarios where outcomes are not guaranteed, such as weather predictions, human behavior, or competitive events [121].